

Mark Scheme — Practice Paper 8 (P1)

M = method mark, A = accuracy mark, B = independent mark. 教师/批改用。

1 $x^2 - 3x + 5 = 2x - 1 \Rightarrow x^2 - 5x + 6 = 0$ (M1). $(x - 2)(x - 3) = 0$ (A1). (2, 3) and (3, 5) (A1). [3]

2 $2x + \frac{\pi}{3}$ lies in $[\frac{\pi}{3}, \frac{7\pi}{3}]$ (B1). $\cos = \frac{1}{2}$ at $\frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$ (M1). $x = 0, \frac{2\pi}{3}, \pi$ (A1 for two, A1 for all three). [4]

3 $(1 + x)^4 = 1 + 4x + 6x^2 + \dots$ (B1); $(2 - x)^5 = 32 - 80x + 80x^2 - \dots$ (B1). x^2 coefficient: $1(80) + 4(-80) + 6(32)$ (M1) $= 80 - 320 + 192 = -48$ (A1). [4]

4(a) Midpoint (3, 2); gradient $AB = 2$, so perpendicular gradient $-\frac{1}{2}$ (M1). $y - 2 = -\frac{1}{2}(x - 3) \Rightarrow x + 2y = 7$ (A1). [2]

4(b) $AB = \sqrt{16 + 64} = 4\sqrt{5}$; height h from C to AB : $\frac{1}{2} \times 4\sqrt{5} \times h = 20 \Rightarrow h = 2\sqrt{5}$ (M1). Move $2\sqrt{5}$ from $M(3, 2)$ along the bisector (direction (2, -1), length $\sqrt{5}$): $C = (3, 2) \pm 2(2, -1)$ (M1). $C = (7, 0)$ or $C = (-1, 4)$ (A1). [Alternative: let $C = (7 - 2t, t)$ and use the shoelace formula.] [3]

5(a) $\frac{20}{2}(a + (-26)) = 50 \Rightarrow a = 31$ (M1 A1). $31 + 19d = -26 \Rightarrow d = -3$ (A1). [3]

5(b) $S_\infty = \frac{3}{1 + \frac{1}{2}} = 2$ (B1). $S_5 = \frac{3(1 - (-\frac{1}{2})^5)}{1 + \frac{1}{2}} = 2(1 + \frac{1}{32}) = \frac{33}{16}$ (M1). Difference $= \frac{33}{16} - 2 = \frac{1}{16}$ (A1). [3]

6(a) $f(x) > 0$ (B2; B1 for $f(x) \geq 0$). [2]

6(b) $x + 2 = \frac{3}{y} \Rightarrow f^{-1}(x) = \frac{3}{x} - 2$ (B1). Domain $x > 0$ (B1). [2]

6(c) $fg(x) = \frac{3}{2x + 1} = 1$ (M1) $\Rightarrow x = 1$ (A1). [2]

7(a) Third quadrant: $\sin \theta = -\frac{3}{5}$, $\cos \theta = -\frac{4}{5}$ (B1 B1; positive values score B1 max). [2]

7(b) $\frac{1 - \sin \theta}{\cos \theta} \times \frac{1 + \sin \theta}{1 + \sin \theta} = \frac{1 - \sin^2 \theta}{\cos \theta(1 + \sin \theta)}$ (M1) $= \frac{\cos^2 \theta}{\cos \theta(1 + \sin \theta)} = \frac{\cos \theta}{1 + \sin \theta}$ (A1 AG). [Cross-multiplying to $1 - \sin^2 \theta = \cos^2 \theta$ is acceptable if presented as an identity chain.] [2]

7(c) $\frac{\cos \theta}{1 + \sin \theta} = 2 \cos \theta \Rightarrow \cos \theta \left(\frac{1}{1 + \sin \theta} - 2 \right) = 0$ (M1). $\cos \theta = 0$ is rejected because the original equation is undefined there; so $1 + \sin \theta = \frac{1}{2} \Rightarrow \sin \theta = -\frac{1}{2}$ (A1). $\theta = 210^\circ, 330^\circ$ (A1). [Including 90° or 270° loses the final A1.] [3]

8(a) $\frac{dy}{dx} = 2x^{-1/2} - 1 = 0 \Rightarrow \sqrt{x} = 2 \Rightarrow x = 4, y = 8 - 4 = 4$ (M1 A1). $\frac{d^2y}{dx^2} = -x^{-3/2} < 0$, maximum at (4, 4) (A1). [3]

8(b) $\frac{2}{\sqrt{x}} - 1 > 0 \Rightarrow \sqrt{x} < 2$ (M1) $\Rightarrow 0 < x < 4$ (A1). [2]

8(c) At $x = 9$: $\frac{dy}{dx} = \frac{2}{3} - 1 = -\frac{1}{3}$. $\frac{dy}{dt} = -\frac{1}{3} \times 0.6$ (M1) $= -0.2$, i.e. y is decreasing at 0.2 units per second (A1). [2]

9(a) $y = \frac{(2x - 1)^3}{3 \times 2} + c = \frac{1}{6}(2x - 1)^3 + c$ (M1 A1). (1, 1): $\frac{1}{6} + c = 1 \Rightarrow c = \frac{5}{6}$ (M1). $y = \frac{1}{6}(2x - 1)^3 + \frac{5}{6}$ (A1). [Expanding first: $y = \frac{4}{3}x^3 - 2x^2 + x + \frac{2}{3}$, equivalent.] [4]

9(b) $4 - x^2 = x + 2 \Rightarrow x^2 + x - 2 = 0 \Rightarrow x = -2, 1$ (B1). Area $= \int_{-2}^1 (4 - x^2 - x - 2) dx = \int_{-2}^1 (2 - x - x^2) dx$ (M1) $= [2x - \frac{1}{2}x^2 - \frac{1}{3}x^3]_{-2}^1$ (A1) $= \frac{7}{6} - (-\frac{10}{3}) = \frac{9}{2}$ (A1). **[4]**

10(a) Angle $AOT = 1.2$ and angle $OAT = 90^\circ$: $\cos 1.2 = \frac{10}{OT}$ (M1) $\Rightarrow OT = \frac{10}{\cos 1.2} = 27.6$ cm (A1). **[2]**

10(b) $AT = 10 \tan 1.2 = 25.72$ (B1). Area of kite $OATB = 2 \times \frac{1}{2} \times 10 \times 25.72 = 257.2$; sector $OAB = \frac{1}{2} \times 100 \times 2.4 = 120$ (M1). Shaded $= 257.2 - 120 = 137$ cm² (A1). **[3]**

10(c) Arc $AB = 10 \times 2.4 = 24$ (B1). Perimeter $= 2 \times 25.72 + 24$ (M1) $= 75.4$ cm (A1). **[3]**

11(a) $k^2 - 4(k + 3) < 0$ (M1) $\Rightarrow k^2 - 4k - 12 < 0 \Rightarrow (k - 6)(k + 2) < 0$ (A1). $-2 < k < 6$ (A1). **[3]**

11(b) $2(x - 1)^2 + 3$ (B1 B1). Range $f(x) \geq 3$ (B1 FT). **[3]**

11(c) $2x^2 - (4 + m)x + 5 = 0$; $(4 + m)^2 - 40 = 0$ (M1 A1). $m = -4 \pm \sqrt{40} = -4 \pm 2\sqrt{10}$ (A1). **[3]**

12(a) Radius $= 3$ (B1). $(x + 2)^2 + (y - 3)^2 = 9$ (B1). **[2]**

12(b) $(x + 2)^2 + (2x + 1)^2 = 9 \Rightarrow 5x^2 + 8x - 4 = 0$ (M1 A1). $(5x - 2)(x + 2) = 0$: $(0.4, 4.8)$ and $(-2, 0)$ (A1). **[3]**

12(c) At $(0.4, 4.8)$: radius gradient $\frac{4.8 - 3}{0.4 + 2} = \frac{3}{4}$ (M1), tangent gradient $-\frac{4}{3}$ (A1). $y - 4.8 = -\frac{4}{3}(x - 0.4) \Rightarrow 4x + 3y = 16$ (A1). **[3]**

Total: 75